

# Inference Overlay Research

## Multi-Cloud Inference Benchmarking & Optimization Platform

A research initiative evaluating production LLM inference across 9 cloud providers — 4 hyperscalers + 5 neo-clouds.

## Research Vision

Modern LLM inference has moved from single-GPU prototypes to multi-cloud production systems, yet the ecosystem lacks standardized benchmarks, cross-cloud normalization, and quantization certification for regulated industries. This initiative addresses these gaps through four complementary research directions.

## Research Directions

| # | Project                                             | Key Takeaway                                                                                 | Venue      |
|---|-----------------------------------------------------|----------------------------------------------------------------------------------------------|------------|
| 1 | <a href="#">InferMark Benchmarking</a>              | 3.2× throughput variance across clouds; 18-61% production overhead nobody measures           | OSDI       |
| 2 | <a href="#">Disaggregated Cross-Cloud Inference</a> | Multi-hop prefill/decode saves 15-29% cost; GQA/MLA are natural fit for cross-cloud disagg   | OSDI/ NSDI |
| 3 | <a href="#">Cross-Cloud Performance Parity</a>      | 4-layer normalization: ±17% → ±4% variance; network topology dominates, not GPU compute      | SysML      |
| 4 | <a href="#">Accuracy-Preserving Quantization</a>    | SSMs most resilient to quantization; MLA latent KV most sensitive; QJL catches 99.97% errors | NeurIPS    |
| — | <a href="#">Full Bucket List (14 ideas)</a>         | Stack-ranked research ideas with takeaways, novelty, and venue fit                           | —          |

## Cloud Providers Under Evaluation

### Hyperscalers

| Provider | GPU SKUs                        | Interconnect        | Regions                   |
|----------|---------------------------------|---------------------|---------------------------|
| AWS      | H100 SXM (p5), B200 (p5e)       | EFA v2 (SRD)        | us-east-1, eu-west-1      |
| GCP      | H100 (A3 Mega), B200 (A3 Ultra) | GPUDirect-TCPX      | us-central1, europe-west4 |
| Azure    | H100 (ND v5), B200 (ND v6)      | InfiniBand NDR      | eastus, westeurope        |
| OCI      | B200 bare-metal (BM.GPU.B200.8) | RDMA Cluster Net v2 | us-ashburn, eu-frankfurt  |

### Neo-Clouds

| Provider      | GPU SKUs       | Interconnect       | Key Differentiator                          |
|---------------|----------------|--------------------|---------------------------------------------|
| CoreWeave     | H100 SXM, B200 | InfiniBand NDR     | GPU-native, K8s-first, fastest availability |
| Lambda        | H100 SXM, B200 | InfiniBand HDR/NDR | Simplest pricing, ML-focused                |
| Together AI   | H100, custom   | Custom fabric      | Inference-optimized, open-source hosting    |
| Crusoe Energy | H100 SXM, B200 | InfiniBand NDR     | Renewable/flare-gas powered, lowest carbon  |
| Voltage Park  | H100 SXM       | InfiniBand NDR     | Large contiguous clusters, HPC-grade        |

## Repository Structure

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```
inference-overlay-research/
├── README.md                # This file
├── research-ideas-bucket-list.md # 14 ideas stack-ranked with takeaways
├── 01-infermark-benchmarking/
│   ├── README.md           # 14-axis benchmarking framework
│   └── architecture.drawio  # System architecture diagram
├── 02-disaggregated-cross-cloud/
│   ├── README.md           # Cross-cloud prefill/decode optimization
│   └── architecture.drawio  # Disaggregation flow diagram
├── 03-cross-cloud-parity/
│   ├── README.md           # 4-layer normalization framework
│   └── architecture.drawio  # Overlay architecture diagram
└── 04-accuracy-preserving-quantization/
    ├── README.md           # TurboQuant certification framework
    └── architecture.drawio  # Evaluation pipeline diagram
```

## License

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